Annotation Guidelines

Variable	Definition	Code
Bibliographic information		
Author(s)	Researchers who conducted and published the study	Open
Title of publication	The title of the study	Open
Year of publication	Year in which the study was published	Open
Type of publication	Journal article or dissertation	1 = journal article 2 = dissertation/thesis 3 = book chapter 4 = conference paper/poster 5 = other
Title of publication venue	Journal in which the study was published	Open
Study design		
Study design	Type of research design used in the study	1 = between-group, experimental (participants are divided into experimental and control groups that receive different treatments) 2 = within-group, quasi-experimental (pretest-post-test design)
Methodology	Type of methodology used in the study	1 = quantitative 2 = mixed (quantitative + qualitative)
Learning context	Environment in which language learning takes place	1 = second language 2 = foreign language 0 = not reported
Region	Region/country in which the research was conducted	Open 0 = not reported
Learner variables		
Participants' age	Mean age or age range of the participants	Open 0 = not reported
Participants' L1	First language of the participants	Open

		0 = not reported 1 = mixed
Participants' L2 (target language)	Language in which outcome measures were elicited	Open
Participants' educational level	Educational level of participants	1 = elementary/primary school 2 = middle school 3 = high school 4 = college/university 5 = not classroom learners 6 = mixed 0 = not reported
Participants' sample size (total)	Number of participants in the study	Open
Experimental group N (Experimental_N)	Number of participants in the experimental group	Open
Control group N (Control_N)	Number of participants in the control group (if any)	Open
L2 proficiency level	Participants' L2 proficiency level	1 = beginner 2 = intermediate 3 = advanced 4 = mixed 0 = not reported/not specified
L2 proficiency justification		0 = not reported 1 = assumed from institutional level 2 = standardized proficiency test (commercially available and developed by testing agencies) 3 = self-reported 4 = other
Participants' language learning experience	Time that participants had been learning the target language (range, mean)	Open 0 = not reported
Sampling	Process of selecting participants	1 = random sampling (each member of the population has an equal chance of being selected) 2 = convenience sampling (non-probability sampling where participants are selected based on their availability and

		willingness to take part in the study) 3 = purposeful sampling (non-probability sampling in which participants are selected based on specific characteristics determined by the researcher) 0 = not reported
Treatment variables		
HiVR treatment setting	Type of environment in which VR tools have been used	1 = classroom 2 = lab 3 = naturalistic (museum, workplace, etc.) 0 = not reported
HMD type	Type of VR headset used in the study	1 = PC-powered VR device (e.g., Oculus Rift, HTC Vive, HTC Vive Pro2) 2 = smartphone-based VR device (e.g., Google Cardboard, Samsung Gear, VR Box, VR Box 2.0) 3 = standalone VR device (e.g., Oculus Go, Oculus Quest, Oculust Quest 2, HTC Vive Focus Plus, DPVR M2 Pro) 0 = not reported
HiVR content type	Type of VR content used in the study	1 = computer-generated (CG) 3D content 2 = 360° videos 3 = 360° videos and computer-generated (CG) 3D content combined 4 = 360° images 5 = 360° images and computer-generated (CG) 3D content combined 0 = not reported
HiVR software	Type of VR application used in the study	1 = previously existing VR content adapted for language learning (e.g., Google Expeditions, Google Tour Creator, EduVenture, Edmersive, CoSpaces, 360° panoramic videos on YouTube)

		2 = off-the-shelf VR applications created for second/foreign language learning (e.g., Mondly VR, House of Languages, ImmerseMe) 3 = custom-made/self-designed VR product (e.g., Unity, Wonda VR, Uptale) 0 = not reported
HiVR learning activities	Type of VR instruction used in the study	1 = personal/self-directed learning (teacher/researcher do not implement specific teaching scenarios for students to follow) 2 = teacher-centered learning (teacher guided students throughout the learning process) 3 = cooperative learning (students are divided into pairs or groups and completed language learning tasks collaboratively) 4 = mixed 0 = not reported
HiVR treatment length (total)	Duration of virtual reality activities	Open 0 = not reported
Number of HiVR treatment sessions		1 = one session 2 = two sessions 3 = three sessions 4 = four sessions 5 = five sessions 6 = six sessions 7 = seven sessions 8 = eight sessions 9 = nine session 0 = not reported
Length of each HiVR treatment session		Open 0 = not reported
HiVR treatment provider		1 = teacher 2 = researcher 3 = teacher/researcher 4 = learner (learner-created VR materials) 5 =mixed 0 = not reported

Control treatment	Type of treatment used in a control group	1 = traditional pedagogical tools, such as textbooks, handouts. 2 = multimedia, which refers to educational materials accessed using videos, images, and/or animation 0 = not reported
Learning outcomes		
Linguistic outcomes	Participants' linguistic gains	1 = vocabulary 2 = reading 3 = listening 4 = writing 5 = speaking 6 = grammar 7 = pronunciation 8 = pragmatics 9 = mixed 0 = not reported
Socio-emotional outcomes	Participants' affective outcomes (e.g., anxiety, motivation, self-efficacy, autonomy, attitude, engagement, enjoyment, ideal L2 self, etc.)	Open
Statistical reporting		
Mean (experimental group, pretest)		(numerical input)
SD (experimental group, pretest)		(numerical input)
Mean (experimental group, posttest)		(numerical input)
SD (experimental group, posttest)		(numerical input)
Mean (control group, pretest)		(numerical input)
SD (control group, pretest)		(numerical input)
Mean (control group, posttest)		(numerical input)
SD (control group, posttest)		(numerical input)
Effect sizes reported		0 = no

	1 = yes
Effect size type	1 = Cohen's d 2 = Hedge's g 3 = eta-squared 4 = partial eta-squared
Effect size value	(numerical input)

Note. Variables without values are continuous, non-categorical, or open-ended.